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Building Energy Management With Reinforcement Learning and Model Predictive Control: A Survey

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ABSTRACT Building energy management has been recognized as of significant importance on improving the overall system efficiency and reducing the greenhouse gas emission. However, the building energy management system is now facing more challenges and uncertainties with the increasing penetration of renewable energy and increasing adoption of different types of electrical appliances and equipment. Classical model predictive control (MPC) has shown effective in building energy management, although it suffers from labour-intensive modelling and complex online control optimization. Recently, with the growing accessibility to building control and automation data, data-driven solutions such as data-driven MPC and reinforcement learning (RL)-based methods have attracted more research interest. However, the potential of integrating these two types of methods and how to choose suitable control algorithms have not been well discussed. In this work, we first present a compact review of the recent advances in data-driven MPC and RL-based control methods for building energy management. Furthermore, the main challenges in these approaches and general discussions on the selection of control methods are discussed.

INDEX TERMS Building energy management, model predictive control, reinforcement learning, data-driven control.

I. INTRODUCTION

The building sector accounts for about 76% of electricity use, 40% of primary energy use and associated greenhouse gas emissions in the U.S. [1], and the similar situation exists in other countries. Therefore, it is essential to reduce energy consumption and carbon emission in buildings to meet national energy and environmental challenges. Furthermore, people spend more than 85% of their time in buildings [2], so well-performing building control methods can also deliver a healthy and comfortable indoor environment for people besides reducing operation costs for buildings.

Recently, the area of building energy management system (BEMS) has gained a significant amount of interest, and the advanced control strategies for BEMS are believed to provide great potential to reduce building energy costs and improve grid energy efficiency and stability [3]. There are two main objectives for BEMS: minimizing energy consumption and ensuring the comfort of the occupants. Comfort management includes the control of the multiple

components of HVAC units with optimized energy consumption. Energy management deals with energy cost optimization by curtailing redundant energy usage and load shifting.

However, building controls in BEMS are becoming complicated, because in addition to traditional services such as lighting and heating, ventilation and air conditioning (HVAC), modern BEMS must respond to on-site intermittent renewables, residential and commercial battery storage units, electric vehicle (EV) charging, and more. All these factors bring increasing challenges and uncertainties in the design of system model. These also boil down to a large-scale complex optimization problem affected by multiple external disturbances, such as variable occupancy conditions and human interaction with the building [4].

The most conventional building control in BEMS is rule-based control (RBC), which relies on some pre-determined schedules to select the setpoint. Its implementation is extremely straightforward while control performance is not optimal. Because for RBC, it's difficult to do the continuous adaptation according to the various environment and most of the times it is also designed without any formal guarantee of constraint satisfaction. Classical model predictive

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control (MPC) uses forecast information to optimize the control inputs and thereby utilizing disturbance predictions for the modelling process, resulting in a reliable control performance. However, several studies have pointed out its limited commercial implementation [5], [6]. This limited adoption of MPC is attributed to predictive model design complexities, as well as increased time constraint and memory footprint required for online optimization [7]. The dependency of the MPC performance on the accuracy of the derived system modeling exponentially increases the computational complexity with the size of the building and the structure of its energy network.

With the growing number of smart meters and sensors installed, building operation data have recently been more accessible through the building automation systems (BAS) [8]. Empowered by large amounts of data, algorithms and significant computing power, lots of studies looked at the various data-driven approaches employed in BEMS. A recent approach is to combine various machine learning (ML) tools with classical MPC to design data-driven MPC strategies that preserve the reliability of classical MPC while reducing the time and computational complexities during online implementation.

Another trend to use data-driven methods in BEMS to enhance control performance is to use reinforcement learning (RL), which has an excellent decision-making capability as a branch of ML. RL refers to a computational approach to learning whereby an agent tries to maximize the total amount of reward it receives while interacting with a complex and uncertain environment. Additionally, RL could leverage the recent and rapid developments in the ML field, such as deep learning and feature encoding, to make better control decisions. RL has been successfully applied in many areas, ranging from gaming [9] to robotics [10]. What's more, it could help users avoid the tedious work of developing and calibrating a detailed model, as is required by MPC; and it can also be used to address consistent implementation issues, namely, time expensive online optimization for advanced control strategies. Adjusting weather and thermal loads, operating chillers at optimal conditions, and controlling blinding devices are some of the applications of RL in this field [6], [11]–[18].

Both of data-driven MPC and RL are effective solutions in BEMS, and may even complement each other with their advantages. Several research efforts have been made to further understand and summarize their implementations. The authors in [8] explore the challenges of data-driven algorithms in building HVAC and energy management applications. An extensive review presented in [19] highlights system model development focusing on building passive thermal mass and data-driven predictive control integration in similar building applications. Two recent reviews [20], [21] focus on pure deep RL-based methods for BEMS, and make a short discussion on choosing RL-based methods according to the data type. In [22] the authors discuss various applications of RL based approaches in BEMS

primarily focusing on the model-free RL algorithms and briefly discuss the overlap between the model-based RL with the traditional MPC methods. To the best of the authors' knowledge, no detailed study has been conducted to give insights into the commonalities and interconnections between these two control methods. As mentioned earlier, traditional MPC and RL approaches can be complementary to each other to achieve improved reliability, data efficiency and energy performance for BEMS. The focus of this survey is to review recent advancements and research challenges in MPC and RL-based methods in BEMS. It also reviews recent developments in RL-based control, including model-based approaches using MPC in modeling and data generation, to enhance data efficiency. The main contributions of this survey are:

- To present a summarized review of recent advances in data-driven MPC and RL-based control methods in BEMS.
- To identify the main challenges and list the connections and differences of these two methods for the BEMS application.
- To provide insight about the pros and cons of each methods and suggest potential directions for engineers and researchers contemplating the use of these methods based on specific control objectives.

The rest of the paper is organized as follows. A brief discussion on the technical background of MPC and RL-based control mechanism along with the building energy management problem in Section II. Section III and Section IV include reviews of articles focusing on data-driven MPC and RL-based building energy management approaches. In Section V, challenges and considerations for the choice of a control approach are discussed, followed by a discussion on future research directions in Section VI.

II. TECHNICAL BACKGROUND

In this section, the typical BEMS architecture and the review of the theoretical background of MPC and RL-based building control strategies are presented.

A. ARCHITECTURE FOR BEMS

A typical BEMS architecture has several important components: residential or commercial buildings, EVs, battery energy storage, renewable sources and the power grid, as illustrated in Fig. 1. The building itself contains household or commercial appliances and equipment, with a smart meter and the control center used to record data and control the equipment in the building. The information flow between the assets and control center is two-way to keep the building working and share information with the external power grid.

Building operation goals commonly deal with the trade-off of ensuring occupants' comfort while minimizing the energy consumption or the associated cost. The metrics for energy optimization mainly focus on net energy consumption such as cost minimization, peak reduction, load shifting, and appliance scheduling. The comfort optimization metrics

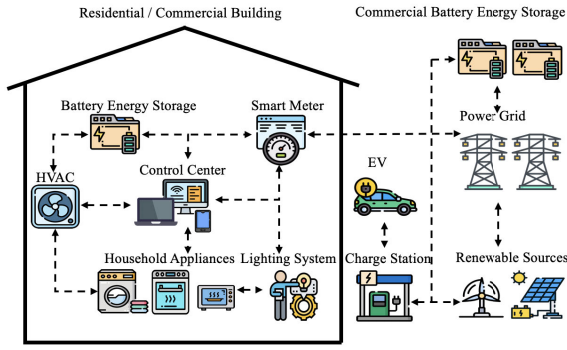


FIGURE 1. Architecture for BEMS. A typical BEMS architecture consists of household or commercial appliances, EVs, battery energy storage units, renewable sources, control center and the power grid.

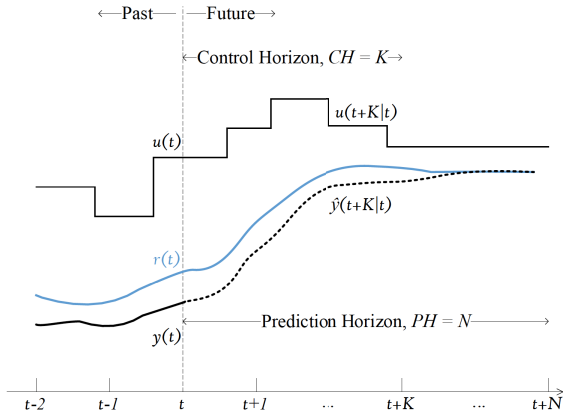


FIGURE 2. Model Predictive Control Strategy. The MPC uses a control-oriented system model to generate output predictions $\hat{y}(\cdot|t)$ for a predefined N time steps in the future. Based on these predictions it optimizes the control inputs for current time step.

focus on keeping control variables like indoor temperature and humidity within a reasonable range.

B. MODEL PREDICTIVE CONTROL

A schematic of a classical MPC approach is shown in Fig. 2 [23] for a tracking control problem. A closed-loop MPC uses a control-oriented system model to generate output predictions, $\hat{y}(\cdot|t)$ for a predefined N time steps in the future, known as the *prediction horizon (PH)*, using the available input-output information at the current time step t . The controller optimizes the manipulated input variables for a shorter interval K , known as the *control horizon (CH)*, keeping the control inputs $u(\cdot|t)$ unchanged during the time interval between $t + K + 1$ and $t + N$, i.e., $\Delta u(\tau|t) = u(\tau|t) - u(\tau - 1|t) = 0, \forall \tau = t + K + 1, \dots, t + N$. The goal is to minimize a problem specific objective function based on the open-loop predictions of controlled outputs. An example of a quadratic MPC objective function tracking a reference trajectory $r(t)$ is given by (1):

$$\min_{u(t+1|t), \dots, u(t+K|t)} \sum_{i=t+1}^N w_y [r(i) - \hat{y}(i)]^2 + w_u [u(i) - u(i-1)]^2, \quad (1)$$

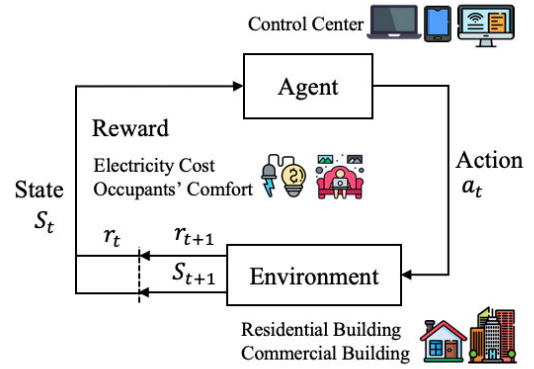


FIGURE 3. A general diagram for reinforcement learning in BEMS. An agent learns to take the optimal set of actions through interaction in a dynamic environment with the goal of maximising a certain reward quantity.

where w_y and w_u are weighting coefficients. Conventionally, only the first value of the optimized control signal, $u(t + 1|t)$ is implemented and the whole optimization is repeated for the next time step by shifting the prediction horizon forward, based on a new set of available information. For this reason, MPC is also known as the receding horizon control [23]. BEMS control is a multi-objective optimization problem. Here, the objective function usually optimizes both comfort and cost objectives. Comfort criteria penalize any deviation from the desired comfort setpoints, which may include indoor temperature, humidity, and overall comfort satisfaction level of the occupants etc. The cost objective, on the other hand, usually manages energy cost, which may incorporate penalties on peak power consumption and electricity expense based on time-of-use rates, etc.

The design of the control-oriented system model is not trivial for building applications and the computational complexity grows with the size of the building. Also, the predictive control optimization at each time step, in a receding horizon setup, is computationally heavy and the order increases exponentially with the length of the prediction horizon as well as the number of control inputs in the optimization problem. The data-driven MPC incorporates ML-based algorithms to navigate through these computational complexities focusing on reducing time and memory footprints required by a classical MPC. As discussed later in Section III, generally in data-driven MPC, the ML algorithms are used either to imitate the control action of a classical MPC during run-time to reduce the online optimization time or to train an ML network to represent the control oriented building model using historical building operation data. The latter may also help in improving the adaptability of a data-driven MPC approach as compared to a classical MPC.

C. REINFORCEMENT LEARNING

In RL, an agent (control center) learns to take the optimal set of actions through interaction in a dynamic environment (a building subject to changing weather conditions, varying grid requirements and occupants with thermal comfort

requirements), with the goal of minimizing/maximizing a certain reward quantity (energy consumption, electricity cost, occupants' comfort, etc.), as shown in Fig. 3. The most common way to model an RL problem is as a Markov Decision Process (MDP). MDP is a discrete-time framework for modelling multi-stage decision making. It can be expressed as a tuple $\langle S, A, P, R, \gamma \rangle$, where S , A , P and R are the sets of states s_t , actions a_t , state transition probabilities p and rewards r ; $\gamma \in [0, 1]$ is a discount factor accounting for future rewards.

RL algorithms can be sorted into model-based and model-free methods based on whether P and R are learned first (model-based) and then used to find the optimal control policy. There are also three types of methods based on what the agent learns in model-free RL: value-based, policy-based and Actor-Critic methods. The State-Action-Reward-State-Action (SARSA) and Q-learning are the most famous classical RL algorithms, which are value-based. They choose the action based on the state value function. Policy-based methods determine the action directly with a parameterized policy function, which could be represented by neural network and updated to increase the likelihood of trajectories that give higher reward. However, the policy methods which calculate the reward through the samples from Monte Carlo methods suffer from the high variance problem in gradient estimates, which give the unbiased but noisy estimate of state value function. To overcome this problem, the Actor-Critic methods parameterize both policy and value functions and simultaneously update them in training. To scale up the RL algorithms with large state and action spaces, deep RL leverages deep neural networks as function approximators to represent policies, reward and value functions. The use of replay buffer and fixed target network also fix the problems of correlations between samples and non-stationary targets in classical RL algorithms [24], [25].

III. REVIEW OF DATA-DRIVEN MPC IN BEMS

ML algorithms are adapted in data-driven MPC design predominantly in two ways. First, a closed-loop MPC is used to generate the input-output data to be used as the training and testing data sets for an ML-based algorithm (Section III-A). Once trained offline, the ML algorithm replaces the MPC during run-time and thereby eliminating the receding horizon control optimization performed by the MPC at each control iteration. However, since the training data is generated using the MPC, the performance of the ML controller is still dependent on the control-oriented building model used by the MPC. Alternatively, another approach (Section III-B) is to use ML-based algorithms to design and train the control-oriented model offline, commonly using closed-loop control measurements from installed RBC or PID controller as the training data. The ML-based control-oriented model parameters are often updated periodically using various adaptive strategies. This control-oriented system model is subsequently used by the MPC for system output predictions during online optimization.

A. MPC GENERATED TRAINING DATA FOR ML-BASED CONTROLLER

In [5] the authors propose an Approximate MPC where deep time delay neural networks (TDNN) and regression tree based regression models are used as approximators for comfort and energy optimization of a multi-zone residential building. The training data is generated using classical closed-loop MPC optimization profiles. Feature engineering is implemented to further reduce model complexity and implementation cost. TDNN-based Approximate MPC reduces the computation time approximately by a factor of 7 as compared to the classical MPC, however it incurs a slight 3% drop in the energy savings. A real-time implementation of a deep-learning-based policy approximator for MPC (DL-MPC) is later presented by [26] in an office building located in Hasselt, Belgium. A deep neural network (DNN)-based microcontroller is presented in [27]. The DNN-based controller has a fast computation time and achieves an 'almost globally optimal solution' of a mixed-integer quadratic program (MIQP).

B. DATA-DRIVEN PREDICTIVE MODEL FOR MPC

An artificial neural network (ANN)-based MPC driven HVAC control system is presented [28]. Radial basic function-based neural networks, are identified by a multi-objective genetic algorithm. The ANN models achieve a suitable trade-off between the accuracy of predicted mean vote (PMV) approximation for comfort and the execution time of the MPC. The discrete MPC uses a branch and bound technique to find the optimal control signal. In [29] the authors present a list of publications, between 2010 to 2016, on the ANN-aided MPC approach in various BEMS applications. In this article, the authors also implemented an ANN-based MPC for a residential building located in Vaughan, Ontario, Canada. The best network after multiple iterations approach has been used to determine the appropriate ANN-driven predictive models for HVAC components. The MPC-driven controller is used in a supervisory level controlling setpoints of local PID controllers. [30] introduces safety aware exploration using model-based deep RL for MPC control-oriented model identification. MPC minimizes energy cost and zone temperature violation using a random-sampling shooting method. The time constraint in real-time implementation is tackled by training an auxiliary policy network that imitates MPC outputs. The proposed method achieves 17.2% to 21.8% reduction in total energy consumption along with $10\times$ reduction in total required training steps as compared to model-free RL using proximal policy optimization (PPO). A novel model-based deep RL algorithm, namely MB²C, is presented in [31], which improves the results presented in [30]. Here, the authors proposed a Model Predictive Path Integral-based control approach instead of a random-sampling shooting method [30] for the MPC optimization since the latter is not efficient in identifying the best action as the randomly chosen batch of action trajectories may not include the best action

sequence. The control-oriented building model, designed for predicting changes in the system states, is identified as an ensemble of multiple adaptive environment-conditioned neural networks (ENNs) which take into account environmental disturbance inputs along with system state and action as network inputs. A weighted ensemble learning approach is adopted to train these ENN models where each model is initialized using random initial network parameters from different batches. MB²C achieves 8.23% more energy savings as compared to the model-based deep RL controller in [30] while maintaining similar comfort. It also achieves higher data efficiency and faster convergence as compared to a model-free PPO-based RL baseline controller as well as the model-based deep RL controller in [30].

A continuous building model adaptation scheme using an ANN-based MPC control-oriented model is proposed in [32]. The adaptive ANN model is updated in each control iteration using the closed-loop MPC data. The ANN-aided MPC has achieved a 58.5% cooling energy reduction along with a 36.7% reduction in electricity consumption while maintaining a comfortable indoor environment. In [33], ANN-based MPC is used for lighting and thermal comfort optimization. A nonlinear autoregressive exogenous model with parallel architecture is used to train the networks that estimate the PMV-based comfort specifications, environmental conditions and power consumption. An input convex neural network (ICNN) quasi-convex MPC is proposed in [34] to ensure input-output convexity of ANN mapping for the control-oriented model. However, ICNN generally only guarantees one-step prediction convexity. In this work, an extension is proposed for multi-shot multi-step prediction convexity with a feed-forward network.

A random forest (RF)-based control-oriented model for MPC is used in [35]. A 29% reduction in HVAC electrical energy consumption and a 63% reduction in thermal energy consumption is achieved. In [36] the authors propose data-predictive control (DPC) where MPC uses regression trees (RT) for prediction. Each RT represents an affine function, that relates the system outputs to the control inputs from the training data, associated with a prediction time step. To alleviate overfitting and high variance of the RTs, an ensemble learning approach is adopted replacing RTs with RFs for each timestep. A variation of the similar approach is presented in [37], where state-space switched affine dynamical linear time-invariant models are identified instead of the affine static prediction. The switched affine state-space models take into account the internal state evolution. Data predictive control based on RFs with affine functions and convex optimization problem for BEMS is presented in [38]. The high dimensionality of the affine function coefficient fitting process is simplified by choosing only two fitted coefficients, implying all past control inputs have similar effects on the state as that of the current control input. Experiment results show better model performance even though the assumption is less realistic as explained by the authors.

In [39], the authors use Gaussian process regression-based MPC control-oriented models to predict heating demand, electric baseload, and natural gas consumption for a gas consumption minimization problem and implemented them at Canmet ENERGY-Varenes, a Natural Resources Canada research facility at Québec, Canada. About 22% reduction in natural gas consumption and greenhouse gas emission is reported, along with 4.3% reduction in the net building heating demand as compared to the current building control operation.

IV. REVIEW OF RL IN BEMS

Both MPC and RL are mainstream solutions to solve control problems. MPC is an optimal control law which relies more on a well understood or modeled transition dynamics with optimization techniques. RL requires a learning process and aims to find the optimal policy under an unknown environment, which focuses on an exploration versus exploitation trade-off.

In this section, recent advances of classical RL, deep RL and methods including MPC in model-based RL for BEMS are reviewed. They show that RL can produce good results through interactions between the agents and building environments.

A. CLASSICAL RL METHODS IN BEMS

A pioneering work in RL-based energy management is proposed by Google DeepMind [40]. Using the RL method, DeepMind AI decrease the electricity bill for cooling a data center by approximately 40%. Then, the authors in [11] formulated the home energy management problem as an MDP problem and then solved with a Q-learning algorithm. A method that combines the tree-like MDP and SARSA algorithm was proposed for appliance scheduling and compared with a Q-learning in [41]. Experimental results show that both SARSA and Q-learning obtain a similar schedule for a finite number of appliances over a 24-hour horizon, but the schedule is arrived at much faster using the variation of SARSA. The authors in [12] compare the value-based approach with the Actor-Critic method for domestic hot water control and find Q-learning performs better.

To combine the users' feedback into the control process and smooth the power consumption profile, the authors in [18] propose an algorithm directly integrating user feedback into its control logic using fuzzy reasoning as reward functions. Then, Q-learning is used to make optimal decisions to schedule the operation of smart home appliances.

However, classical RL methods suffer severely from the curse of dimensionality when using historical data or applied to continuous operations in building control. There has been some effort to reduce the dimension in training data. The authors in [42] use a deep neural network-based dimension reduction technique, then compress the previous ten indoor temperatures and control signals into six hidden states. In their following work [13], the predicted states are used to help improve the performance of the RL controller as MPC,

and they find that including weather forecasts as states could improve the performance by 27%.

B. DEEP RL METHODS IN BEMS

In continuous control and discrete control problems which have large action spaces and state spaces, the curse of dimensionality hinders the implementation of RL in practice. So, deep neural networks have been widely used recently as function approximators to represent policies, rewards, and value functions in building control to scale up the RL algorithms. Deep Q-learning has been adopted to optimize the operation cost or energy consumption of a single building HVAC system through EnergyPlus tool in [14], where the system determines discrete air flow rate based on time, zone temperature and environment disturbances. The authors in [43] substitute the Q-table with an ANN that maps current and target temperature directly to their Q-values, allowing the controller to work with continuous states and actions, and also to speed up the learning process, and experiments results demonstrate that the deep RL-based algorithm is more effective in energy cost reduction compared with the traditional rule-based approach. The authors in [14] implement the deep RL hot water controllers in 32 Dutch houses by taking into account occupant interaction and hot water system dynamics. They find that, compared with the fixed schedule or fixed setpoint control, the RL controller reduces energy consumption by almost 20% while maintaining occupant comfort.

Except for controlling single device in building, there also exist some methods consider the control operations of heterogeneous home appliances and distributed energy resources according to the consumer's comfort and preferences. The authors in [14] propose a two-level hierarchical deep RL-based energy management framework for BEMS to handle the interdependent operation between the home appliances at the first level and the energy storage system (ESS)/EV at the second level. Then the optimal policy for charging and discharging actions of the ESS and EV is independently determined considering the energy consumption schedule of aggregated home appliances. The approach employs an actor-critic method where the controllable home appliances are scheduled at the first level. The energy-saving system and EVs are scheduled at the second level to cover the aggregated washing machine (WM) and air conditioner (AC) loads, which are calculated at the first level along with the fixed load of the uncontrollable appliances. A DQN-based home energy management that considers both the appliances scheduling and EV charging scheduling is presented in [14]. To deal with both discrete and continuous actions to jointly optimize the schedules of all kinds of appliances, a deep RL approach based on trust region policy optimization is proposed in [44]. The approach considers three kinds of appliances including deferrable appliances, regulatable appliances, and critical appliances in the simulation model and directly learns from raw observation data of the appliance states, real-time electricity price, and outdoor temperature.

Apart from the BEMS in a residential home, there are many existing works on building control with consideration of commercial building. The authors in [15] propose a deep RL-based framework for efficiently controlling four-building energy subsystems so that the total energy consumed by all subsystems can be minimized while still maintaining user comfort.

Another way to deal with multiple appliances or multiple buildings control problems is to use multi-agent RL (MARL), where each agent corresponded to various home appliances/buildings can communicate with each other. In classical RL algorithms and deep RL works, they often investigate energy policies for household appliances under same environments and reward setting, thereby restricting the algorithm's effectiveness and generalization in real scenarios. MARL requires setting several agents, where different types of household appliances represent different agents with their actions and rewards. The authors in [45] propose a methods for the optimal scheduling of different household appliances to optimize energy utilization. The authors in [16] propose a MARL algorithm to minimize HVAC energy consumption without sacrificing user comfort by adjusting both the building and chiller set-points. To speed up the training process, they use transfer learning in which the agents are trained on sub-sets of HVAC systems and the learned network weights are used to initialize the multiple agents. Furthermore, an hour-ahead deep RL algorithm for BEMS based on multi-agent RL is proposed in [46], which optimizes both shiftable appliances and AC considering the uncertainty in future prices. The consumption scheduling problem is first formulated as a finite MDP (FMDP) with discrete time steps. Then, the FMDP is utilized to model the hour-ahead energy consumption scheduling problem to minimize the electricity bill, as well as DR, including dissatisfaction. The authors in [16] focus on the large-scale BEMS optimization problem for smart homes and propose a collective MARL algorithm with continuous action space to achieve flexible and precise control. Apart from MARL, multi-objective learning also received research attention considering different control objectives in BEMS. A multi-objective algorithm is proposed based on human appliances interaction, which considers scheduling in the context of energy consumption and discomfort level of the home user [17].

However, some works also point out that it is impractical to let the RL agent explore the state space fully in a real building environment, because an unacceptably high economic cost may be incurred when the controller makes bad actions in exploring phase. Plus, it may take a long time for the deep RL agent to learn an optimal policy if trained in a real-world environment. To reduce the dependency on a real building environment, many model-based deep RL control methods have been developed and some research has tried to incorporate the domain knowledge into the training process or suggests using the MPC method to enhance the understanding of the dynamic process. The authors in [47] use the observed data in EnergyPlus to develop a building energy model,

and then use the model as the environment simulator to train the deep RL agent off-line based on the asynchronous advantage actor critic algorithm. In this way, the deep RL agent's potentially harmful exploration in real-world HVAC is limited. In [48], long short-term memory (LSTM) is used to build the environment model using historical data, in which the inputs of LSTM models are the current state and action, while their outputs are the next state and reward. Then the agent is trained using deep deterministic policy gradient. The authors in [6] encode the domain knowledge on planning and linear system dynamics into the RL controller by differentiable MPC policy. The system uses offline pre-training by imitating the existing controller and online learning to interact with the environment and update its policy. The results show that it can save 16.7% of cooling demand compared to the manual set-point controller. What also deserves to be mentioned is that training an RL controller is data and time-demanding. To accelerate the training process, the approach proposed in [18] works with a single agent and uses a reduced number of state-action pairs as rewards functions. The authors in [49] uses fuzzy rules to discretize continuous states-actions and to reduce dimensions in a smart residential community. The approach proposed in [50] uses Gaussian Process Regression (GPR) to compress six states into two.

V. DISCUSSIONS ON THE MPC-BASED METHODS AND RL-BASED METHODS

This section focuses on the prevalent challenges associated with the real-time implementation of the MPC and RL-driven control methods in different BEMS applications. Also, some considerations on the choices of control methods to address these challenges are discussed.

A. CHALLENGES OF MPC-BASED METHODS FOR BEMS

Challenges of classical MPC for BEMS applications, concerning model dependency and computational complexity in iterative optimization, are broadly acknowledged in the available literature. With the recent development of smart communities, shared energy storage, and an increasing number of EVs participating in the grid-based demand-side management, the complexity of the BEMS is growing rapidly. Although the design specifications of modern buildings are more accessible nowadays, old building retrofits pose a hindrance in control-oriented model identification for MPC design. Eventually, this limits the possibilities of fast adaptation of classical MPC. It is worth mentioning that the data-driven MPCs, reviewed in III-A, also suffer from a similar drawback. Even though they perform well to resolve the time-expensive nature of the classical MPC optimization routines, these data-driven MPCs are restricted to specific buildings when trained on data generated by the 'teacher' MPCs.

Recent fundamental research works in advanced MPC are addressing these challenges. For example, Explicit MPC [51] precomputes the control law offline using piecewise affine functions and thereby completely eliminates the online optimization requirements by equivalently expressing the

MPC as a lookup table of linear gains. Introducing ML in such a method is proposed in [52] for hybrid systems. A nonlinear state-space control-oriented model identification method for MPC is presented in [53] using ML techniques based on autoencoders and neural networks. Preference-based RL method is used for semi-automated MPC calibration in [54], [55]. The current research trend shows a promising path forward in developing feasible embedded real-time controllers for BEMS applications using data-driven MPC strategies.

B. CHALLENGES OF RL-BASED METHODS FOR BEMS

With the fast progress of reinforcement learning, RL-based BEMS has recently drawn a lot of attention. However, there are still several challenges to using RL, especially deep RL, for real-world applications, e.g., building energy management. Here, we summarize the main challenges that should be addressed to further improve the RL-based BEMS solution. The foremost challenge is **data efficiency** [56], [57]. For most of the current RL-based approaches, the learning agent needs to interact with the environment for a large number of steps to learn a reliable control policy. Second is the **safety concern** of RL-based methods [58], [59]. The RL-based method may output some control actions that bring the system into some undesirable system states. The safety concerns are largely neglected in most of the current RL-based solutions. The third one is the **generalization** [22], [47]. In the real world, human behaviour varies from day to day, which indicates that the corresponding optimal control policies will be different. Thus, to further reduce the operation cost of the building, we need to design a solution that can generalize to different consumption scenarios. The fourth one is **learning from historical data** [22], [60]. Most of the current RL-based BEMS methods rely on an accurate simulator, which can be costly to develop. Thus, it is important to investigate how to learn a BEMS control policy with historical data directly.

C. ADVANTAGES COMPARISON AND CHOICE OF MPC-BASED AND RL-BASED METHODS

Although many challenges exist, data-driven MPC and RL-based methods are believed to be the most powerful strategies to handle decision-making problems in BEMS. Based on specific control objectives, both strategies can be well designed, selected and even combined to get the desired control performance. A comparison of these two methods based on some key features can be summarized as follows:

- 1) **Control strategy:** RL explicitly considers the whole problem of a goal-directed agent interacting with an uncertain environment. MPC treats modelling and planning as two separate tasks. The quality of the model is evaluated by criteria such as prediction error, which may affect control performance. Data-driven MPC approaches adapt their models online, based on observations from the system, which is similar to model-based RL with planning methods. This allows

the controller to improve over time, given limited prior knowledge of the system.

- 2) **Data used for control design:** RL is based on experience, which is used to reduce the need for iterative methods. Moreover, depending on the formulation of the problem and the richness of experience data, the chances of convergence are high. MPC and model-based RL which use planning, on the other hand, integrates forecast information and considers future disturbances to handle multiple constraints and objectives.
- 3) **Length of time horizon:** MPC optimizes finite-length trajectories based on a pre-specified or learned system model. RL-based algorithms can find an infinite horizon optimal policy under unknown dynamics of the system, based on interactions and external cost signals (where system dynamics could also be learned in model-based RL).

Considerations for the choice of control methods in real-world applications can be made based on the above features. MPC in optimal control assumes a well understood or modelled transition dynamics. This is typically the case for problems that are traditionally studied in control engineering. When the detailed mechanistic modelling of transition dynamics is almost impossible for a complex systems, we could leverage RL to see how the system behaves or start learning the transition dynamics from experiments and data. Also, MPC is desirable as it prevents the accumulation of model error by planning in a finite horizon into the future. This ensures that each plan does not need to be perfect as re-planning guarantees improvement.

On the other hand, methods could also be chosen based on a characterization of the current research challenges and opportunities in data-driven MPC and RL-based methods in BEMS, which are computational complexity, data efficiency, safety and robust adaptability. The ML-based controllers trained using classical MPC (Section III-A) reduce online computational time as compared to the classical MPC, by eliminating the necessity of online control optimization at each time step. The main advantage of RL-based methods is its ability to continuously adapt over time according to dynamic environments, which also results in low data efficiency and long training periods problems. This could be accomplished by different exploration processes such as setting the initial learning phase or offering available prior knowledge. MPC with a data-driven predictive model (Section III-B) also preserves the reliability ensured by the classical MPC, while simplifying the predictive model identification process. Safety constraints can be easily added to an MPC problem formulation to ensure reliability in the control performance, while in RL, the agents needs to learn how to avoid bad actions through trial and errors.

VI. CONCLUSION

Building energy management is of significant importance to improve the overall efficiency of power systems. In this

paper, a compact survey of recent developments for data-driven model predictive control (MPC) and reinforcement learning (RL) RL-based control algorithms for BEMS has been presented. Data-driven MPC faces the challenges of design complexity and time-consuming computations, while RL-based methods face the data-efficiency, safety and robust adaptability problems.

Considerations for the choice of a control methods in real-world applications have been presented from the perspectives of the method's features and remaining research challenges. The straightforward way to choose the model is that if transition dynamics are known or can be modeled easily, then a traditional MPC technique may be chosen as there is a rich literature available on this topic. If modelling becomes difficult, one may resort to a hybrid RL and optimal control methods to search for an optimal policy from the interaction with the environment. Moreover, combining the classical MPC with RL-based prediction approaches in model-based RL appears to offer a suitable trade-off between reliability and practicality of implementation. For MPC, the data-driven approaches have contributed to improving computational complexity, time constraints, adaptability and simplification of the control-oriented model design which is an essential part of predictive control. For RL-based BEMS control, MPC can contribute to ensuring data-efficiency, safety and robustness. Relatively simpler data-driven predictive models combined with robust control strategies seem to chart a reasonable path forward in order to achieve desired time-efficiency, reliability and adaptability for real-time building energy management.

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REFERENCES

- [1] U. S. Department of Energy. (Jan. 4, 2015). *Quadrennial Technology Review 2015: Chapter 5-Increasing Efficiency of Buildings Systems and Technologies*. Accessed: Feb. 25, 2021. [Online]. Available: <https://www.energy.gov/downloads/chapter-5-increasing-efficiencybuildings-systems-and-technologies>
- [2] N. E. Klepeis, W. C. Nelson, W. R. Ott, J. P. Robinson, A. M. Tsang, P. Switzer, J. V. Behar, S. C. Hern, and W. H. Engelmann, "The national human activity pattern survey (NHAPS): A resource for assessing exposure to environmental pollutants," *J. Exposure Anal. Environ. Epidemiol.*, vol. 11, no. 3, pp. 231–252, 2001.
- [3] R. Missaoui, H. Joumaa, S. Ploix, and S. Bacha, "Managing energy smart homes according to energy prices: Analysis of a building energy management system," *Energy Buildings*, vol. 71, pp. 155–167, Mar. 2014.
- [4] D. Mariano-Hernández, L. Hernández-Callejo, A. Zorita-Lamadrid, O. Duque-Pérez, and F. S. García, "A review of strategies for building energy management system: Model predictive control, demand side management, optimization, and fault detect & diagnosis," *J. Building Eng.*, vol. 33, Jan. 2021, Art. no. 101692.
- [5] J. Drgoňa, D. Picard, M. Kvasnica, and L. Helsen, "Approximate model predictive building control via machine learning," *Appl. Energy*, vol. 218, pp. 199–216, May 2018. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0306261918302903>
- [6] B. Chen, Z. Cai, and M. Bergés, "Gnu-RL: A precocious reinforcement learning solution for building HVAC control using a differentiable MPC policy," in *Proc. 6th ACM Int. Conf. Syst. Energy-Efficient Buildings, Cities, Transp.*, Nov. 2019, pp. 316–325.

- [7] J. Cígler, D. Gyalistras, J. Široky, V. Tiet, and L. Ferkl, "Beyond theory: The challenge of implementing model predictive control in buildings," in *Proc. 11th Rehva World Congr., Clima*, vol. 250, 2013, pp. 1–10.
- [8] E. T. Maddalena, Y. Lian, and C. N. Jones, "Data-driven methods for building control—A review and promising future directions," *Control Eng. Pract.*, vol. 95, Feb. 2020, Art. no. 104211. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0967066119301832>
- [9] D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis, "A general reinforcement learning algorithm that masters chess, shogi, and go through self-play," *Science*, vol. 362, no. 6419, pp. 1140–1144, 2018.
- [10] S. Levine, P. Pastor, A. Krizhevsky, J. Ibarz, and D. Quillen, "Learning hand-eye coordination for robotic grasping with deep learning and large-scale data collection," *Int. J. Robot. Res.*, vol. 37, nos. 4–5, pp. 421–436, 2017.
- [11] Z. Wen, D. O'Neill, and H. Maei, "Optimal demand response using device-based reinforcement learning," *IEEE Trans. Smart Grid*, vol. 6, no. 5, pp. 2312–2324, Sep. 2015.
- [12] K. Al-jabery, Z. Xu, W. Yu, D. C. Wunsch, J. Xiong, and Y. Shi, "Demand-side management of domestic electric water heaters using approximate dynamic programming," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 36, no. 5, pp. 775–788, May 2017.
- [13] F. Ruelens, B. J. Claessens, S. Vandaal, B. De Schutter, R. Babuska, and R. Belmans, "Residential demand response of thermostatically controlled loads using batch reinforcement learning," *IEEE Trans. Smart Grid*, vol. 8, no. 5, pp. 2149–2159, Sep. 2017.
- [14] T. Wei, Y. Wang, and Q. Zhu, "Deep reinforcement learning for building HVAC control," in *Proc. 54th Annu. Design Autom. Conf.*, Jun. 2017, pp. 1–6.
- [15] X. Ding, W. Du, and A. Cerpa, "OCTOPUS: Deep reinforcement learning for holistic smart building control," in *Proc. 6th ACM Int. Conf. Syst. Energy-Efficient Buildings, Cities, Transp.*, Nov. 2019, pp. 326–335.
- [16] S. Nagarathinam, V. Menon, A. Vasan, and A. Sivasubramaniam, "MARCO—Multi-agent reinforcement learning based control of building HVAC systems," in *Proc. 11th ACM Int. Conf. Future Energy Syst.*, Jun. 2020, pp. 57–67.
- [17] M. Diyan, B. N. Silva, and K. Han, "A multi-objective approach for optimal energy management in smart home using the reinforcement learning," *Sensors*, vol. 20, no. 12, p. 3450, Jun. 2020.
- [18] F. Alfaverh, M. Denai, and Y. Sun, "Demand response strategy based on reinforcement learning and fuzzy reasoning for home energy management," *IEEE Access*, vol. 8, pp. 39310–39321, 2020.
- [19] A. Kathirgamanathan, M. De Rosa, E. Mangina, and D. P. Finn, "Data-driven predictive control for unlocking building energy flexibility: A review," *Renew. Sustain. Energy Rev.*, vol. 135, Jan. 2021, Art. no. 110120. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S1364032120304111>
- [20] L. Yu, S. Qin, M. Zhang, C. Shen, T. Jiang, and X. Guan, "A review of deep reinforcement learning for smart building energy management," 2020, *arXiv:2008.05074*.
- [21] H. Zhang, D. Wu, and B. Boulet, "A review of recent advances on reinforcement learning for smart home energy management," in *Proc. IEEE Electr. Power Energy Conf. (EPEC)*, Nov. 2020, pp. 1–6.
- [22] Z. Wang and T. Hong, "Reinforcement learning for building controls: The opportunities and challenges," *Appl. Energy*, vol. 269, Jul. 2020, Art. no. 115036. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0306261920305481>
- [23] E. F. Camacho and C. B. Alba, *Model Predictive Control*. London, U.K.: Springer, 2007.
- [24] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. H. A. Graves, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [25] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, "Continuous control with deep reinforcement learning," 2015, *arXiv:1509.02971*.
- [26] J. Drgoňa, L. Helsen, and D. Vrabie, "Stripping off the implementation complexity of physics-based model predictive control for buildings via deep learning," in *Proc. 33rd Conf. Neural Inf. Process. Syst.*, 2019, pp. 1–7. [Online]. Available: <https://www.climatechange.ai/papers/neurips2019/34>
- [27] B. Karg and S. Lucia, "Deep learning-based embedded mixed-integer model predictive control," in *Proc. Eur. Control Conf. (ECC)*, Jun. 2018, pp. 2075–2080.
- [28] P. M. Ferreira, A. E. Ruano, S. Silva, and E. Z. E. Conceição, "Neural networks based predictive control for thermal comfort and energy savings in public buildings," *Energy Buildings*, vol. 55, pp. 238–251, Dec. 2012. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S037877881200388X>
- [29] A. Afram, F. Janabi-Sharifi, A. S. Fung, and K. Raahemifar, "Artificial neural network (ANN) based model predictive control (MPC) and optimization of HVAC systems: A state of the art review and case study of a residential HVAC system," *Energy Buildings*, vol. 141, pp. 96–113, Apr. 2017. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0378778816310799>
- [30] C. Zhang, S. R. Kuppannagari, R. Kannan, and V. K. Prasanna, "Building HVAC scheduling using reinforcement learning via neural network based model approximation," in *Proc. 6th ACM Int. Conf. Syst. Energy-Efficient Buildings, Cities, Transp. (BuildSys)*. New York, NY, USA: ACM, Nov. 2019, pp. 287–296, doi: [10.1145/3360322.3360861](https://doi.org/10.1145/3360322.3360861).
- [31] X. Ding, W. Du, and A. E. Cerpa, "MB2C: Model-based deep reinforcement learning for multi-zone building control," in *Proc. 7th ACM Int. Conf. Syst. Energy-Efficient Buildings, Cities, Transp. (BuildSys)*. New York, NY, USA: ACM, Nov. 2020, pp. 50–59, doi: [10.1145/3408308.3427986](https://doi.org/10.1145/3408308.3427986).
- [32] S. Yang, M. P. Wan, W. Chen, B. F. Ng, and S. Dubey, "Model predictive control with adaptive machine-learning-based model for building energy efficiency and comfort optimization," *Appl. Energy*, vol. 271, Aug. 2020, Art. no. 115147. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0306261920306590>
- [33] R. Eini and S. Abdelwahed, "A neural network-based model predictive control approach for buildings comfort management," in *Proc. IEEE Int. Smart Cities Conf. (ISC2)*, Sep. 2020, pp. 1–7.
- [34] F. Bünning, A. Schalbeter, A. Aboudonia, M. H. de Badyn, P. Heer, and J. Lygeros, "Input convex neural networks for building MPC," 2020, *arXiv:2011.13227*.
- [35] T. Hilliard, L. Swan, and Z. Qin, "Experimental implementation of whole building MPC with zone based thermal comfort adjustments," *Building Environ.*, vol. 125, pp. 326–338, Nov. 2017. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0360132317304110>
- [36] A. Jain, F. Smarra, and R. Mangharam, "Data predictive control using regression trees and ensemble learning," in *Proc. IEEE 56th Annu. Conf. Decis. Control (CDC)*, Dec. 2017, pp. 4446–4451.
- [37] F. Smarra, A. Jain, R. Mangharam, and A. D'Innocenzo, "Data-driven switched affine modeling for model predictive control," *IFAC-PapersOnLine*, vol. 51, no. 16, pp. 199–204, 2018. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S2405896318311509>
- [38] F. Bünning, B. Huber, P. Heer, A. Aboudonia, and J. Lygeros, "Experimental demonstration of data predictive control for energy optimization and thermal comfort in buildings," *Energy Buildings*, vol. 211, Mar. 2020, Art. no. 109792. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0378778819320638>
- [39] N. Cotrufo, E. Saloux, J. M. Hardy, J. A. Candanedo, and R. Platon, "A practical artificial intelligence-based approach for predictive control in commercial and institutional buildings," *Energy Buildings*, vol. 206, Jan. 2020, Art. no. 109563. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0378778819322431>
- [40] DeepMind. (Jul. 20, 2016). *DeepMind AI Reduces Google Data Centre Cooling Bill by 40%*. [Online]. Available: <https://deepmind.com/blog/article/deepmind-ai-reduces-google-datacentre-cooling-bill-40>
- [41] N. Chauhan, N. Choudhary, and K. George, "A comparison of reinforcement learning based approaches to appliance scheduling," in *Proc. 2nd Int. Conf. Contemp. Comput. Informat. (IC3I)*, Dec. 2016, pp. 253–258.
- [42] F. Ruelens, S. Iacovella, B. Claessens, and R. Belmans, "Learning agent for a heat-pump thermostat with a set-back strategy using model-free reinforcement learning," *Energies*, vol. 8, no. 8, pp. 8300–8318, Aug. 2015.
- [43] J. Vázquez-Canteli, J. Kämpf, and Z. Nagy, "Balancing comfort and energy consumption of a heat pump using batch reinforcement learning with fitted Q-iteration," *Energy Proc.*, vol. 122, pp. 415–420, Sep. 2017.
- [44] H. Li, Z. Wan, and H. He, "Real-time residential demand response," *IEEE Trans. Smart Grid*, vol. 11, no. 5, pp. 4144–4154, Sep. 2020.
- [45] S. Lee and D.-H. Choi, "Reinforcement learning-based energy management of smart home with rooftop solar photovoltaic system, energy storage system, and home appliances," *Sensors*, vol. 19, no. 18, p. 3937, Sep. 2019.

- [46] R. Lu, S. H. Hong, and M. Yu, "Demand response for home energy management using reinforcement learning and artificial neural network," *IEEE Trans. Smart Grid*, vol. 10, no. 6, pp. 6629–6639, Nov. 2019.
- [47] Z. Zhang, A. Chong, Y. Pan, C. Zhang, and K. P. Lam, "Whole building energy model for HVAC optimal control: A practical framework based on deep reinforcement learning," *Energy Buildings*, vol. 199, pp. 472–490, Sep. 2019.
- [48] Z. Zou, X. Yu, and S. Ergen, "Towards optimal control of air handling units using deep reinforcement learning and recurrent neural network," *Building Environ.*, vol. 168, Jan. 2020, Art. no. 106535.
- [49] S. Zhou, Z. Hu, W. Gu, M. Jiang, and X. Zhang, "Artificial intelligence based smart energy community management: A reinforcement learning approach," *CSEE J. Power Energy Syst.*, vol. 5, no. 1, pp. 1–10, 2019.
- [50] Y. R. Yoon and H. J. Moon, "Performance based thermal comfort control (PTCC) using deep reinforcement learning for space cooling," *Energy Buildings*, vol. 203, Nov. 2019, Art. no. 109420.
- [51] A. Bemporad, *Explicit Model Predictive Control*. London, U.K.: Springer, 2019, pp. 1–7, doi: [10.1007/978-1-4471-5102-9_10-2](https://doi.org/10.1007/978-1-4471-5102-9_10-2).
- [52] D. Masti, T. Pippia, A. Bemporad, and B. D. Schutter, "Learning approximate semi-explicit hybrid MPC with an application to microgrids," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 5207–5212, 2020.
- [53] D. Masti and A. Bemporad, "Learning nonlinear state-space models using autoencoders," *Automatica*, vol. 129, Jul. 2021, Art. no. 109666. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0005109821001862>
- [54] A. Bemporad and D. Piga, "Global optimization based on active preference learning with radial basis functions," *Mach. Learn.*, vol. 110, no. 2, pp. 417–448, Feb. 2021.
- [55] M. Zhu, A. Bemporad, and D. Piga, "Preference-based MPC calibration," in *Proc. Eur. Control Conf. (ECC)*, Jun. 2021, pp. 638–645.
- [56] X. Xu, Y. Jia, Y. Xu, Z. Xu, S. Chai, and C. S. Lai, "A multi-agent reinforcement learning-based data-driven method for home energy management," *IEEE Trans. Smart Grid*, vol. 11, no. 4, pp. 3201–3211, Jul. 2020.
- [57] B. Mbuwir, F. Ruelens, F. Spiessens, and G. Deconinck, "Battery energy management in a microgrid using batch reinforcement learning," *Energies*, vol. 10, no. 11, p. 1846, Nov. 2017.
- [58] R. Harper, "Inside the smart home: Ideas, possibilities and methods," in *Inside the Smart Home*. London, U.K.: Springer, 2003, pp. 1–13.
- [59] H. Li, Z. Wan, and H. He, "Constrained EV charging scheduling based on safe deep reinforcement learning," *IEEE Trans. Smart Grid*, vol. 11, no. 3, pp. 2427–2439, May 2020.
- [60] H. Berlink and A. H. Costa, "Batch reinforcement learning for smart home energy management," in *Proc. 24th Int. Joint Conf. Artif. Intell.*, Jun. 2015, pp. 1–7.



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